JWST Constraints on BH – Host Relation at 5 < z < 6.5

Olivier Gilbert, Jinyi Yang, Feige Wang, Bingcheng Jin, Xiangyu Jin, Hyunseop Choi, and the COSMOS-3D collaboration

Background

The relation between supermassive black hole (SMBH) mass and host-galaxy stellar mass provides key evidence for the co-evolution of black holes and galaxies. While this relation is well established in the local Universe, its form at early cosmic times remains poorly constrained.

At z>5, only a limited number of broad-line AGN (BLAGN) host galaxies have been studied, preventing robust measurements of the SMBH–host relation. Determining whether high-redshift systems follow the local relation is essential for understanding the growth of the first SMBHs and galaxies.

Recently discovered “Little Red Dots” (LRD) are compact red objects mainly found at high-z, which exhibit a “V-shaped” spectral energy distribution (SED) and seem to contain SMBH.

Data/Sample

The BLAGN sample is selected from a JWST wide field survey: COSMOS-3D. It combines multi-band photometry with slitless spectroscopy, which enables us to select bright, red, and compact targets with emission lines that we visually inspect to select a pure broad-line Hα sample.

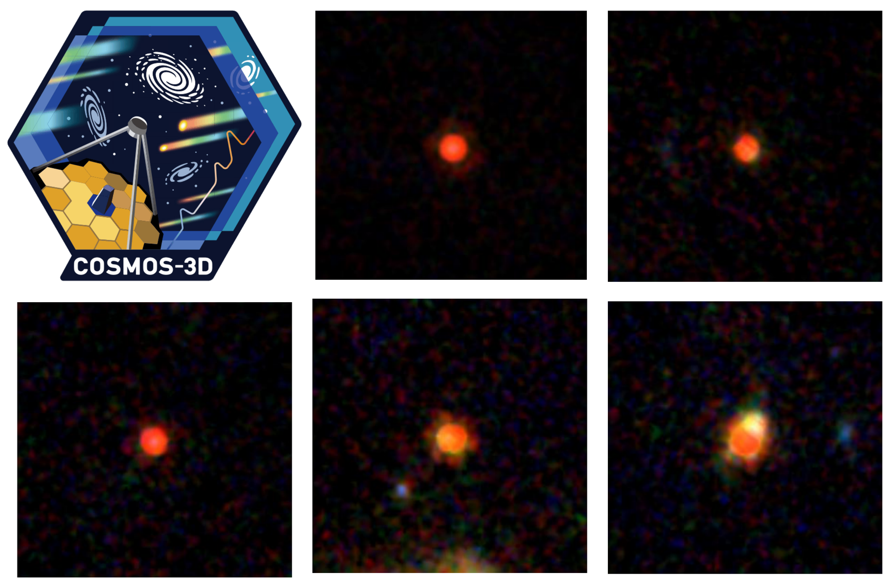
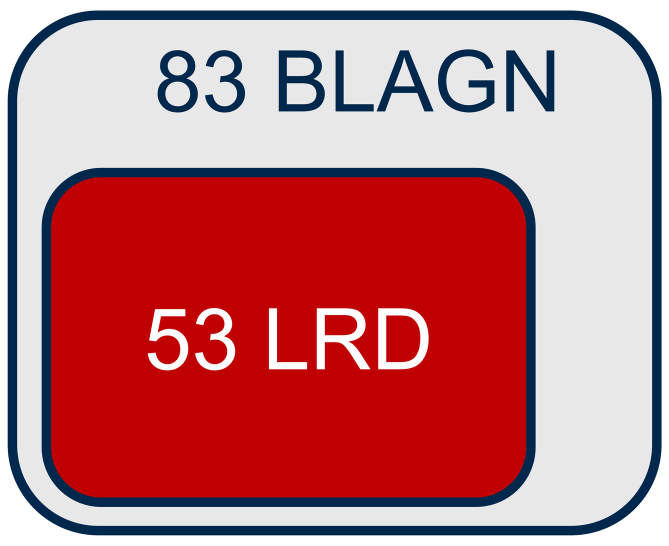


Figure 1. NIRCam RGB example cutouts of the BLAGN sample along with the JWST COSMOS-3D survey logo.



Out of 83 BLAGN in our sample, 53 satisfy a standard LRD color selection.

Spectral fitting

The NIRCam/WFSS F444W spectra is used to fit the Hα broad-line with narrow+broad components and obtain the line properties. The black hole masses are then calculated using

$$M_{\rm BH} = 2\times10^{6}\left(\frac{L_{\rm H\alpha}}{10^{42}\ {\rm ergs\ s^{-1}}}\right)^{0.55}\left(\frac{\rm FWHM_{H\alpha}}{10^{3}\ {\rm km\ s^{-1}}}\right)^{2.06} M_\odot$$

Some BLAGN exhibit Balmer absorption, indicative of very dense gas. This is a rare phenomenon at low-z (<0.1% of AGN), but more common at high-z (10-20% of BLAGN). The origin of the absorption is still debated.

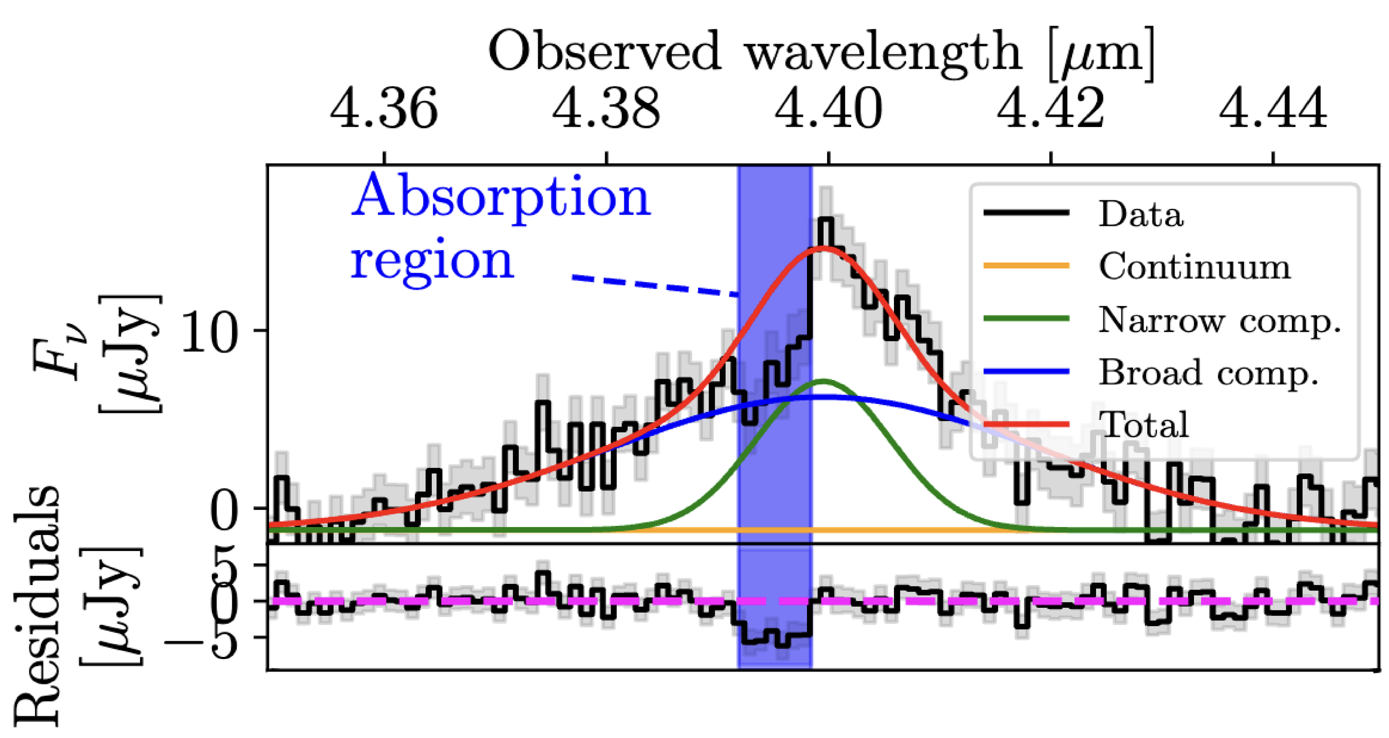


Figure 2. Spectral modelling example of a BLAGN exhibiting Balmer absorption.

Online version

Spatial Decomposition

To isolate the flux from the host galaxy, we perform a 2D spatial decomposition using GALIGHT. We assume a Sérsic ellipse profile for the host and a simple scaled point-source for the AGN. This fitting is performed in all six NIRCam bands (F115W,F150W,F200W,F277W,F356W,F444W).

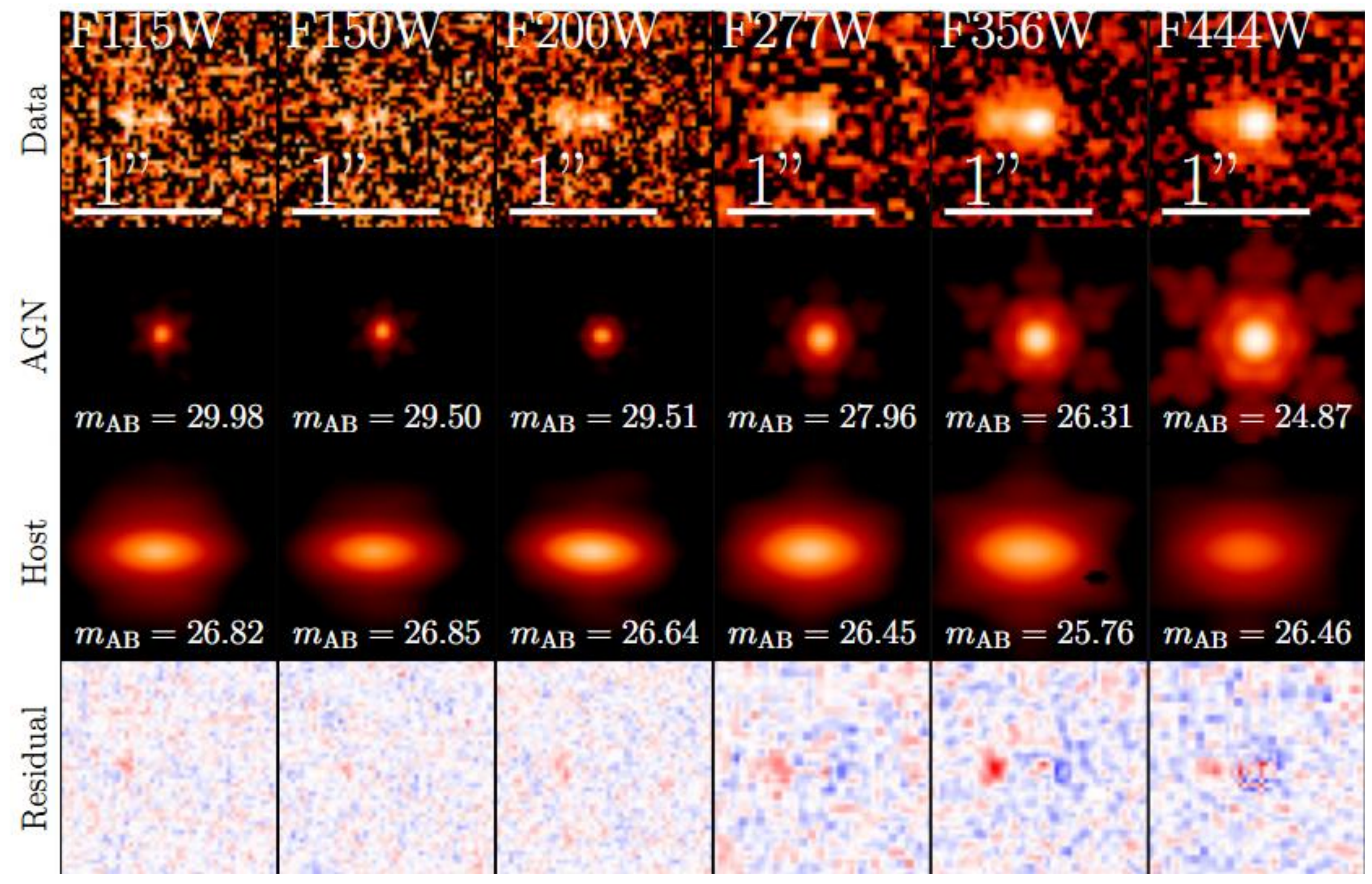


Figure 3. AGN+Host spatial decomposition example for the six NIRCam filters of the same target.

SED Fitting

To constrain the stellar mass, the host galaxy photometry from the decomposition is fitted using BAGPIPES with a delayed tau formation history.

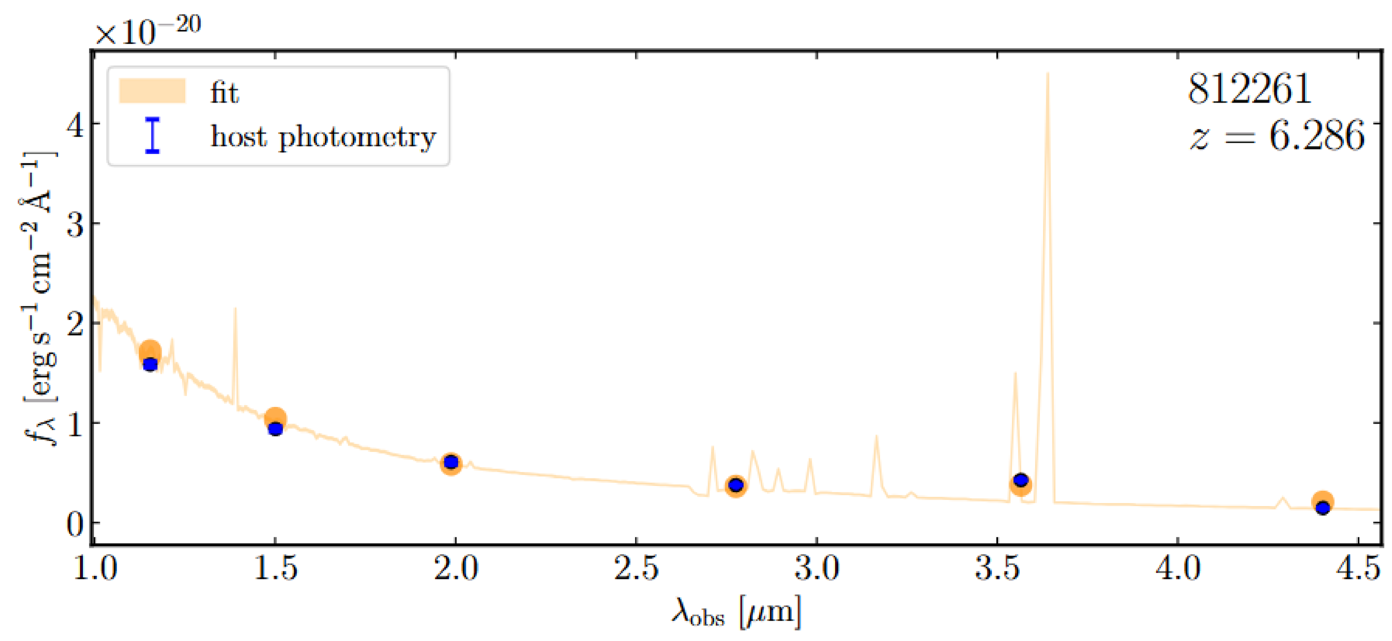


Figure 4. BAGPIPES SED modelling example for a BLAGN.

Results

- Host galaxies contribute significantly to the rest-frame UV flux, but the AGN dominates in the rest-frame optical.
- High-z BLAGN show a large scatter in the $M_{\rm BH} - M_\star$ plot, showing the local relation isn’t formed yet, or has a much larger scatter. The BLAGN also showcase higher $M_{\rm BH}/M_\star$ ratio.
- Both Balmer-absorption BLAGN and LRD have more massive BH than regular BLAGN and there is a large overlap between the two subsamples.

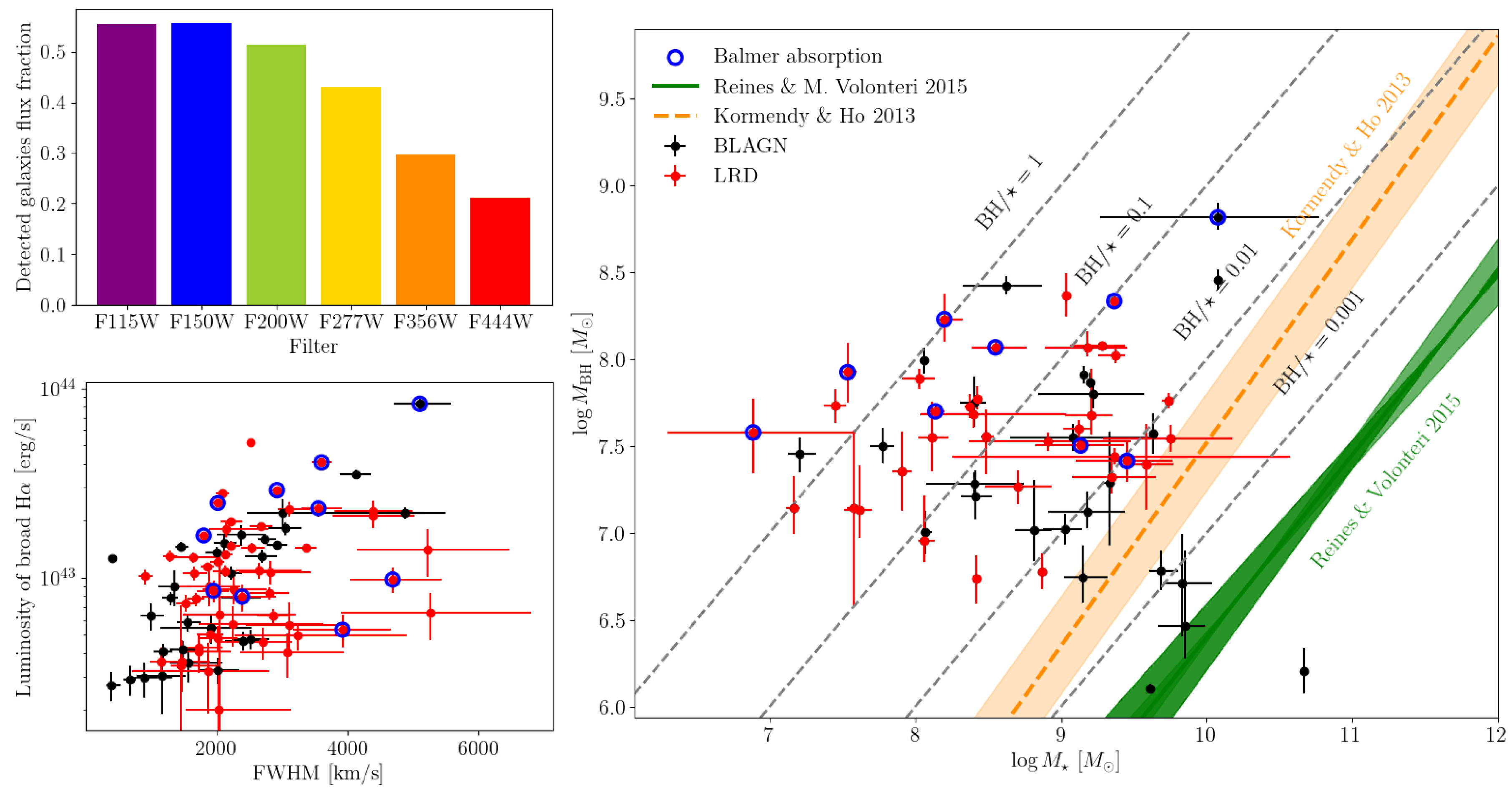


Figure 5. Top left: Host galaxy flux contribution for the different filters, showing a decrease towards redder wavelengths. Bottom left: Spectral properties used to measure the BH mass. Right: Black hole mass – Stellar mass relation for the BLAGN sample.